Vo Van Tu Tai

Software Engineer (Backend & Frontend) · Node.js, TypeScript & React · E-Commerce & Loyalty Systems

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PROFESSIONAL SUMMARY

Information Systems Graduate (IUH, GPA 3.26/4.0 · Completed all degree requirements · Available immediately for full-time) with hands-on full-stack development capabilities bridging **Backend (Node.js, NestJS, Express.js, TypeScript)** and **Frontend (React.js, Modern JavaScript ES6+, Responsive UI)**. Solid background in designing RESTful APIs (**OpenAPI 3.0**), architecting normalized relational databases (**PostgreSQL, MySQL 3NF**), and implementing concurrency controls (pessimistic locking, Redis distributed locks SETNX). Practical experience engineering real-world E-Commerce platforms, transactional carts, asynchronous message queues (**RabbitMQ**), and multi-container Dockerization. Active practitioner of **AI-Driven Engineering**, leveraging **Claude Code, Cursor, and GitHub Copilot** to accelerate learning, author test suites, and write clean, maintainable code. Highly motivated to contribute to **TAPTAP's E-Commerce & Loyalty systems** across both backend services and frontend applications.

CORE TECHNICAL COMPETENCIES

Programming Languages: TypeScript, JavaScript (ES6+), PHP (OOP, MVC Architecture), Python, Java (Core OOP), SQL, HTML5, Modern CSS3.

Frontend Engineering: React.js, Responsive Web Design, Modern Vanilla CSS, Single Page Applications (SPA), DOM Manipulation, State Management, Fetch API / Axios.

Backend & API Architecture: Node.js (NestJS, Express.js), PHP MVC, RESTful API Design (OpenAPI 3.0, Postman), Idempotent APIs (`Idempotency-Key`), Asynchronous Webhooks (HMAC-SHA512).

Databases & Caching: PostgreSQL (3NF, B-Tree Indexing), MySQL (ACID Transactions, Foreign Keys), MongoDB, Redis (In-memory caching, Distributed Locks SETNX, Geospatial Queries).

Distributed Systems & Concurrency: RabbitMQ Message Broker (Decoupled queues, burst traffic buffering), State Machine Workflows, Concurrency Control (`SELECT ... FOR UPDATE`, TTL hold).

DevOps, Cloud & Tools: Docker, Docker Compose, Git / GitHub (Feature branching, PR review workflows), Linux / Bash fundamentals, CI/CD pipeline basics, Jira, Figma.

AI-Driven Engineering & Practices: Claude Code, GitHub Copilot, Cursor, ChatGPT (Rapid learning, test authoring, schema synthesis), Clean Code, Agile/Scrum.

WORK EXPERIENCE

HomeNest Software — Ho Chi Minh City, Vietnam

Business Analyst · PM Assistant (Engineering Liaison) — *Mar 2026 – Sep 2026*

- **Technical Specification & Schema Alignment:** Collaborated closely with Lead Dev and UI/UX to translate multi-service business logic (On-demand home services, carpooling, marketplace) into precise technical specifications, JSON data models, and testable acceptance criteria (Given-When-Then), preventing schema mismatches before sprint kickoff.
- **State-Machine Architecture (Interactive Bidding):** Architected a state-driven price negotiation module (propose ↔ counter-bid ↔ accept/reject with TTL expiration); specified strict status locking and race-condition guards to guide backend developers in implementation.
- **API & Performance Cost Optimization:** Resolved high third-party map API budget consumption and mobile battery drain by specifying an event-driven GPS refresh checkpoint mechanism (triggered strictly on order pickup/completion instead of continuous polling), drastically saving external API invocation costs while ensuring defensible audit trails.
- **QA Testing & UAT Coordination:** Executed hands-on functional and smoke testing on staging builds using Postman and web clients; conducted defect triage in Jira, coordinating rapid bug resolution with developers across 2+ release cycles.

J&T Express — Ho Chi Minh City, Vietnam

Operations Process Coordinator (Full-time) — *Sep 2025 – Feb 2026*

- **SQL Data Reconciliation & System Integrity:** Formulated daily SQL queries to cross-verify discrepancies between physical branch delivery manifests and enterprise logistics system logs; prepared daily SQL data integrity reports for management.
- **Pipeline Throughput Optimization:** Identified throughput bottlenecks handling high-volume daily shipments; restructured shift processing into a closed-loop unidirectional sorting flow (Intake → Weigh → Scan → Dispatch), substantially accelerating turnaround times.
- **Standard Operating Procedures (SOP):** Authored standard operational exception procedures categorizing delivery anomalies, establishing clear boundary checks and cross-shift operational accountability.

FEATURED TECHNICAL PROJECTS

Nexus Express — Logistics OMS & Distributed Workflow Engine [Live Demo ↗] [GitHub ↗] Distributed Architecture Project

NestJS, TypeScript, PostgreSQL (3NF), RabbitMQ, React, OpenAPI 3.0, Camunda BPMN 2.0, Docker *Sep 2025 – Feb 2026*

- **Full-Stack Microservices & Live Tracking UI:** Built an enterprise Order Management System with live tracking deployed at **tracking.nexus-ex.site**; implemented **React** web client consuming **NestJS (TypeScript)** APIs and **RabbitMQ** event queues for seamless real-time parcel updates without UI freezing.
- **Idempotent API Contracts & Camunda BPMN:** Designed OpenAPI 3.0 specifications enforcing `Idempotency–Key` headers on scan endpoints to prevent duplicate records upon retries; modeled parcel lifecycles with Camunda BPMN 2.0 to enforce strict transition validation gates.
- **PostgreSQL 3NF Data Modeling & Audit Trails:** Designed normalized schema with composite B-Tree indexes on tracking codes and timestamps; maintained tamper-proof event audit logs enabling 100% discrepancy reconciliation during transit audits.

Cab Booking & Real-Time Dispatch Platform [GitHub ↗] High-Concurrency Distributed Project

Node.js, Express.js, React Client, Redis (In-Memory Locks & Caching), Leaflet / Geolocation, Docker *Dec 2025 – Jan 2026*

- **High-Concurrency Dispatch with Redis Locks:** Engineered driver-rider matching backend utilizing **Redis distributed locks (SETNX with TTL)**, completely eliminating simultaneous ride acceptance race conditions when multiple drivers accept the same trip.
- **Interactive Geolocation UI & Sub-Millisecond Queries:** Integrated interactive map tracking with Leaflet and Redis geospatial indexes (`GEOADD`, `GEORADIUS`) to render real-time driver coordinates with sub-millisecond proximity recalculations.
- **State Machine Resiliency & Containerization:** Modeled resilient ride state machine (`Requested → Accepted → Arrived → In–Trip → Completed / Cancelled`) with dynamic pricing; containerized services into Docker Compose with health checks.

Song Tai Shop — Full-Stack E-Commerce Platform [GitHub ↗] Web Application Project

PHP (OOP, Custom MVC Framework), MySQL (3NF & ACID), JavaScript (Fetch API/AJAX), Docker Compose *Nov 2025 – Jan 2026*

- **Custom MVC Architecture & Security:** Engineered a native PHP MVC framework from scratch utilizing a Front-Controller (`index.php`), custom regex routing for REST-like methods, PDO prepared statements eliminating SQL injection, CSRF token validation, XSS sanitization, and session-based RBAC.
- **ACID Transactional Cart & Concurrency:** Protected inventory integrity during simultaneous checkouts by wrapping order placement and stock deduction in strict MySQL ACID transactions (`START TRANSACTION`, `COMMIT`, `ROLLBACK`) with pessimistic row locking (`SELECT ... FOR UPDATE`).
- **Dynamic Client UI & Containerization:** Developed dynamic shopping cart interactions using JavaScript Fetch API for real-time quantity recalculation without page reloads; packaged the full LAMP stack (PHP-FPM, Apache, MySQL) into **Docker Compose** for reproducible zero-friction local deployments.

Tourist Accommodation Booking & Hospitality Platform [GitHub ↗] Web Application Project

PHP (OOP & MVC Architecture), MySQL (3NF & Indexing), JavaScript, VNPay IPN Webhook, RESTful APIs, Docker *Oct 2025 – Dec 2025*

- **Full-Stack Booking Engine & RBAC:** Architected an end-to-end multi-tenant booking platform with RESTful APIs for catalog search, date-range availability checks, and multi-tier pricing; enforced granular RBAC isolating Guests, Hosts, and Super-Admins.
- **Concurrency Defense & Overbooking Lock (TTL Hold):** Solved critical room reservation race conditions during peak traffic spikes through a stateful two-phase booking hold: temporarily locks rooms in `Pending` state with a 15-minute TTL hold, auto-releasing inventory via timeout daemon if unconfirmed.
- **Cryptographic Payment Gateway & Indexing:** Integrated VNPay payment gateway using HMAC-SHA512 checksum validation with idempotent IPN webhook processing; optimized MySQL 3NF database with composite indexes on `(property_id, check_in, check_out, status)` for sub-second query latency.

EDUCATION

Industrial University of Ho Chi Minh City (IUH) Ho Chi Minh City, Vietnam

Bachelor of Engineering (B.Eng.) in Information Systems · GPA: 3.26 / 4.0 *Sep 2022 – 2026 (Completed All Degree Requirements · Available Full-time)*

- **Relevant Coursework:** Software Engineering, Object-Oriented Programming (OOP), Data Structures & Algorithms, Systems Analysis & Design, Database Management Systems (Advanced SQL Queries & Indexing), Web Application Development, Service-Oriented Architecture (SOA), Enterprise Resource Planning (ERP - Odoo), Artificial Intelligence & Intelligent Systems Concepts. (*Official Academic Transcript available upon request*).
- **Languages & Professional Traits:** English (Professional Working Proficiency — capable of reading English technical documentation, API specifications & developer discussions), Vietnamese (Native), proactive problem-solving, and rapid adaptation to modern engineering workflows.